

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404932
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kondratiuk; Jens et al.

Sealing geometry for a mobile power tool

Abstract

A sealing geometry for a mobile power tool is provided wherein the mobile power tool has a drive unit, wherein the sealing geometry is designed to seal off an interior (IB), containing a lubricant, of the drive unit from an exterior (AB) outside the drive unit. The sealing geometry has a seal that is contactless at least during operation of the drive unit.

Inventors:	Kondratiuk; Jens (Buchs, CH), Hummel; Daniel (Buchs, CH)
Applicant:	Hilti Aktiengesellschaft (Schaan, LI)
Family ID:	73037737
Assignee:	Hilti Aktiengesellschaft (Schaan, LI)
Appl. No.:	18/032074
Filed (or PCT Filed):	October 14, 2021
PCT No.:	PCT/EP2021/078455
PCT Pub. No.:	WO2022/089945
PCT Pub. Date:	May 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230407970 A1	Dec. 21, 2023

Foreign Application Priority Data

EP	20204286	Oct. 28, 2020
----	----------	---------------

Publication Classification

Int. Cl.: F16J15/3204 (20160101); **B25D16/00** (20060101); **B25D17/26** (20060101); **F16J15/40** (20060101); **F16J15/447** (20060101)

U.S. Cl.:

CPC F16J15/3204 (20130101); **B25D17/26** (20130101); **F16J15/40** (20130101); **F16J15/447** (20130101); B25D16/00 (20130101); B25D2250/365 (20130101)

Field of Classification Search

CPC: F16J (15/3204); F16J (15/40); F16J (15/447); F16J (15/441); F16J (15/3272); B25D (16/00); B25D (17/26); B25D (2250/365)

USPC: 173/197

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2478649	12/1948	Wightman	277/377	F16J 15/42
3266811	12/1965	Knauth	277/433	F16J 15/164
4448426	12/1983	Jackowski	277/568	F16J 15/3256
4709930	12/1986	Forch	277/430	F16J 15/3244
4852890	12/1988	Borowski	277/377	F16J 15/164
4928979	12/1989	Nagasawa	277/424	F16J 15/3264
4968044	12/1989	Petrak	384/140	F16J 15/3456
4974860	12/1989	Anzue	277/377	F16J 15/164
5201529	12/1992	Heinzen	277/411	F16J 15/3256
5350181	12/1993	Horve	277/560	F16J 15/3244
6170832	12/2000	Ernst	277/412	F16C 33/763
6170834	12/2000	Vogt	277/571	F16J 15/3256
6354598	12/2001	Huang	277/572	F16J 15/3244
6367807	12/2001	Rockwood	277/412	F16J 15/4478
6474653	12/2001	Hintenlang	277/572	F16J 15/3256
7066467	12/2005	Sakata	277/423	F16C 19/49
7464942	12/2007	Madigan	277/569	F16J 15/3252
7926815	12/2010	Niebling	277/571	F16J 15/3264

8186689	12/2011	Miyazawa	277/423	B25F 5/02
8246253	12/2011	Masui	384/477	F16J 15/3456
8276917	12/2011	Nakagawa	277/423	F16J 15/324
8356941	12/2012	Mason	384/477	F16C 33/80
8474825	12/2012	Nakagawa	277/572	F16J 15/164
8534674	12/2012	Maskaliunas	277/551	F16J 15/3208
8809243	12/2013	Schmid- Amelunxen et al.	N/A	N/A
9039013	12/2014	Artiles et al.	N/A	N/A
9568105	12/2016	Oshima	N/A	F16J 15/3244
9618117	12/2016	Nakagawa	N/A	F16J 5/3256
10054229	12/2017	Organisciak	N/A	F16C 33/7889
10288175	12/2018	Schmitt	N/A	F16J 15/3256
10955006	12/2020	Laplante	N/A	F16J 15/3264
11085004	12/2020	Seemeyer et al.	N/A	N/A
2003/0006563	12/2002	Cater	277/549	F16J 15/3456
2003/0184021	12/2002	Hatch	277/395	F16J 15/3228
2004/0070150	12/2003	Chitren	277/358	F16J 15/4474
2004/0113366	12/2003	Mause	277/394	F16C 33/7859
2007/0080503	12/2006	Soda	277/551	F16J 15/3256
2008/0122182	12/2007	Parker	29/402.02	F16J 15/3488
2009/0001671	12/2008	Dietle	277/559	F16J 15/164
2010/0102515	12/2009	Miyazawa et al.	N/A	N/A
2010/0194054	12/2009	Nishigaki	277/549	F16J 15/3288
2012/0007316	12/2011	Terasawa	277/361	F16C 33/7886
2012/0149616	12/2011	Schmid- Amelunxen et al.	N/A	N/A
2012/0280458	12/2011	Artiles et al.	N/A	N/A
2012/0305279	12/2011	Schadow et al.	N/A	N/A
2013/0035267	12/2012	Schweigkofler et al.	N/A	N/A
2015/0211639	12/2014	Grimanis	277/303	F16J 15/447
2016/0108966	12/2015	Krebs et al.	N/A	N/A
2016/0153561	12/2015	Blechschiidt	277/553	F16J 15/3232

2017/0268679	12/2016	Itadani	N/A	F16J 15/3412
2017/0284533	12/2016	Biermann	N/A	F16J 15/3204
2018/0106374	12/2017	Itadani	N/A	F16J 15/3448
2018/0347702	12/2017	Sakano	N/A	F16J 15/164
2020/0080647	12/2019	Specht	N/A	F16J 15/164
2020/0157454	12/2019	Seemeyer et al.	N/A	N/A
2020/0166138	12/2019	Yonai	N/A	F16J 15/3264
2021/0148466	12/2020	Von Fragstein	N/A	C08J 7/123
2023/0407970	12/2022	Kondratiuk	N/A	F16J 15/447

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3330473	12/1983	DE	N/A
3438755	12/1985	DE	N/A
3734888	12/1988	DE	N/A
199 34 170	12/2000	DE	N/A
199 59 472	12/2000	DE	N/A
102 06 823	12/2002	DE	N/A
199 50 705	12/2008	DE	N/A
10 2008 029 994	12/2009	DE	N/A
10 2013 200 117	12/2013	DE	N/A
10 2013 208 203	12/2013	DE	N/A
10 2014 214 049	12/2015	DE	N/A
10 2016 208 258	12/2016	DE	N/A
1 0550 700	12/2004	EP	N/A
2 180 214	12/2009	EP	N/A
2 520 835	12/2013	EP	N/A
2011/026576	12/2010	WO	N/A
WO 2011/082880	12/2010	WO	N/A
2011/131331	12/2010	WO	N/A
WO 2013/060464	12/2012	WO	N/A
2015/144342	12/2014	WO	N/A
2016/131543	12/2015	WO	N/A
2018/206252	12/2017	WO	N/A

OTHER PUBLICATIONS

European Patent Office, International Search Report and Written Opinion in International Patent Application No. PCT/EP2021/078455, mailed on Jan. 24, 2022. cited by applicant
Hasslinger, “Hydro Lubricants: Are water-containing lubricants the future?”, URL: <https://www.fluid.de/hydraulik/hydro-lubricants-ist-wasser-im-schmirstoff-die-zukunft-208.html>, published Oct. 15, 2017, accessed Mar. 17, 2021. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

- (1) This patent application is the U.S. National Stage of International Patent Application No. PCT/EP2021/078455, filed Oct. 14, 2021, which claims the benefit of European Patent Application No. 20204286.7, filed Oct. 28, 2020, which are each incorporated by reference.
- (2) The present invention relates to a sealing geometry for a mobile power tool, for example a hand-held power tool, in particular for working on natural or artificial, in particular cementitious, rocks, wherein the mobile power tool has a drive unit, wherein the sealing geometry is designed to seal off an interior, containing a lubricant, of the drive unit from an exterior outside the drive unit. The invention also relates to a mobile power tool.
- (3) Parts, moving relative to one another, of mobile power tools, in particular of hand-held power tools, and in particular moving parts of the drive unit, are frequently subject to significant friction. The friction causes considerable losses of work output of the hand-held power tool. Substantial temperature increases can also occur locally or over the entire hand-held power tool. These in turn frequently cause accelerated wear. These temperature increases also represent a safety hazard for a user of the hand-held power tool. To reduce this friction, there is usually a lubricant in the drive unit. A sealing geometry then serves to seal off the interior, comprising lubricant, from the exterior. In particular in the case of powerful hand-held power tools, the sealing geometry is subject to significant mechanical and thermal stress. This is associated with high maintenance costs and relatively short service lives.
- (4) Therefore, it is desirable, and thus an object of the present invention, to present a generic sealing geometry and a mobile power tool which have particularly long service lives.
- (5) Surprisingly, the object is achieved by a sealing geometry for a mobile power tool, in particular for working on natural or artificial, in particular cementitious, rocks, wherein the mobile power tool has a drive unit, wherein the sealing geometry is designed to seal off an interior, containing a lubricant, of the drive unit from an exterior outside the drive unit, wherein the sealing geometry has a seal that is contactless at least during operation of the drive unit.
- (6) The mobile power tool can be or comprise a hand-held power tool. The mobile power tool can also be or comprise a construction robot, in particular designed for working in civil engineering above and/or below ground.
- (7) A seal that is contactless at least during operation of the drive unit can be understood as being a seal that is designed to provide a seal between two elements that are spaced apart, as a rule slightly, from one another, wherein the seal is spaced apart from at least one of the two elements at least during operation of the drive unit. It can thus be configured not to be in contact with this element at least during operation of the drive unit.
- (8) The invention is based on the idea that a contactless seal during operation of the drive unit is subject to no or at least no appreciable friction. Mechanical and associated thermal stresses on the sealing geometry can thus be reduced. As a result, the service life of the sealing geometry can be considerably increased. The maintenance costs required for the sealing geometry can also be considerably reduced or even avoided entirely. Temperatures at and/or in a mobile power tool in which the sealing geometry is used can also be reduced. Lubricant cooling devices can have smaller dimensions or be dispensed with entirely. The work output and/or, for example in battery-operated devices, the operating life of the mobile power tool can also be increased or lengthened on account of the reduced friction.
- (9) The sealing geometry can particularly preferably be designed to be used with an aqueous

lubricant. To this end, it can be adapted in particular to properties and demands that arise on account of the high water content, for example be configured to be water- and/or water-vapor-resistant.

(10) The aqueous lubricant can comprise at least 5 percent, preferably at least 15 percent, water. In particularly preferred embodiments of the invention, the lubricant has a water content of between 25 and 40 percent, in particular of 28 or of 38 percent. Water can form the main constituent of the lubricant. The aqueous lubricant can be fluid, in particular liquid, at a normal operating temperature of the mobile power tool. The water content of the lubricant may be at most 70 percent. The lubricant may comprise, in addition to water, at least one additive, in particular an antiwear additive, an anticorrosion additive and/or an antimicrobial, growth-inhibiting additive. The lubricant may also comprise at least one glycol. The glycol or glycols can, in particular after water, make up a second largest share of the lubricant.

(11) The indication of a percentage can preferably be understood to be a weight-related percentage, that is to say a percentage by weight.

(12) The sealing geometry can have a sealing lip that bears against a mating surface of the mobile power tool, in particular of the drive unit, when the drive unit is at a standstill, in order to seal off the interior from the exterior. In this way, a durable seal can be ensured when the drive unit is at a standstill.

(13) The or a sealing lip of the sealing geometry may be configured such that, during operation of the drive unit, this sealing lip comes away from the or a mating surface or that, during operation of the drive unit, a pressure force of this sealing lip on this mating surface is reduced at least compared with when the drive unit is at a standstill. The sealing geometry may be designed such that the coming away or the reduction in the pressure force arises only from a predefined, in particular lower, limit value, for example a minimum rotational speed of a shaft of the mobile power tool. The sealing geometry can in particular be designed such that the coming away or the reduction in the pressure force arises as a result of a centripetal force brought about by the operation of the drive unit.

(14) The sealing geometry may have a seal carrier on and/or in which the sealing lip is arranged, wherein preferably the sealing lip fits in an indentation in the seal carrier. The sealing lip can then be made from a different material than the seal carrier. For example, the sealing lip can be made from a thermoplastic. The seal carrier can be made from an elastomer. It is also conceivable for the sealing lip to be made from an elastomer. The seal carrier can also be made from a thermoplastic.

(15) The sealing lip may have a V-shaped or an at least substantially V-shaped cross section. It may in particular have a sealing lip wing. The sealing lip wing may form an arm of the V-shaped or substantially V-shaped cross section. As a result of the V-shape, the sealing lip, in particular when it is made from an elastic material, can have a low spring constant.

(16) The sealing geometry may be designed to create at least one vortex flow in the lubricant during operation of the drive unit, said vortex flow improving the sealing action of the sealing geometry. In particular, the vortex flow can create a suction effect such that only a reduced amount of lubricant can even reach the sealing lip. As a result of the vortex flow, an additional sealing action can be achieved, in particular through a further increase in the flow resistance in the exit direction. For example, when the sealing geometry comprises a region of two cylindrical or at least substantially cylindrical parts, in the case of which preferably one of the parts is arranged in the other part, such vortex flows can be generated for example by Taylor-Couette flows.

(17) To this end, the seal carrier of the sealing geometry can alternatively or additionally have at least one vortex structure for creating the at least one vortex flow. The vortex structure can be formed on a surface portion of the seal carrier. It can be configured in the form of an elliptic arc, in particular of a circular arc, in cross section.

(18) It is also conceivable for the sealing geometry to be designed to have and/or to create a sealing fluid, in particular sealing air, at least during operation of the drive unit. The sealing fluid, in

particular the sealing air, can to this end form an air cushion blocking the lubricant.

(19) The sealing geometry can also be configured in a ferromagnetically sealing manner.

(20) The sealing geometry may have a lubricant labyrinth seal. This can be formed upstream of the sealing lip along a flow path of the lubricant. The sealing geometry may have a constriction. The constriction may be elongate. The sealing geometry may have at least one flow obstacle. In general, the sealing geometry can be configured to increase a flow resistance of the lubricant in an exit direction from the interior to the exterior. It can have for example a Tesla valve or at least a Tesla-valve-like structure. It can in particular be configured such that, at least during operation of the drive unit, the lubricant builds up a counterpressure to further lubricant flowing up in the exit direction.

(21) It is also conceivable for at least one thread to be formed in the region of the lubricant labyrinth seal, in particular in the region of the constriction. By way of the thread, lubricant can additionally be conveyed away from the sealing lip during operation of the drive unit. To this end, the thread can preferably be formed so as to extend in a radial direction or at least so as not to extend axially with respect to the shaft.

(22) In a particularly cost-effective embodiment of the mobile power tool, the sealing geometry may comprise an acrylonitrile butadiene (in the following: NBR). The NBR may be a hydrated NBR or an NBR modified in some other way. Thus, the sealing geometry can be resistant to water, hot water and/or water vapor.

(23) The sealing geometry may also comprise a fluorine-containing material. In particular, it may comprise a fluorine-containing rubber. The sealing geometry may be made from a particularly temperature-resistant elastomer. The fluorine-containing rubber may particularly preferably be a fluororubber (in the following: FKM). Alternatively or additionally, it is also conceivable for the fluorine-containing rubber to be or to comprise a perfluororubber (FFKM). In general, the fluorine-containing rubber may be a fluorinated elastomer. It is also conceivable for the fluorine-containing material to be a polytetrafluoroethylene (PTFE). The sealing geometry may also comprise different fluorine-containing materials, in particular different fluorine-containing rubbers and/or a PTFE.

(24) The sealing geometry is usable in a particularly flexible manner, in particular also usable in conjunction with an aqueous lubricant, when the sealing geometry comprises a water-vapor-resistant material. Preferably, the sealing geometry can be made from at least one water-vapor-resistant material.

(25) The categorization of a material as water-vapor-resistant, hot-water-resistant and/or water-vapor resistant can be based on an examination pursuant to ISO 1817. Preferably, the categorization may relate to testing at 60° C. and/or, in particular for testing the water-vapor resistance, at 100° C. The duration of the testing taken as a basis can be at least 72 hours, particularly preferably 168 hours and/or 1008 hours. Testing twice over 168 hours and over 1008 hours is very particularly preferred. A change in tensile strength and/or elongation at break can be used as a categorization feature. In particular, upon testing over 168 hours at 60° C., a particularly suitable material can exhibit a reduction in tensile strength of at most 14%, particularly preferably of at most 10% and very particularly preferably of at most 8%.

(26) The possible uses of the sealing geometry according to the invention can also be expanded when the sealing geometry comprises a high-temperature-resistant material. Preferably, the sealing geometry may be made from a high-temperature-resistant material. In particular, the sealing geometry may comprise a material that is sustainably resistant to temperatures of at least 120° C., particularly preferably of at least 150° C. The sealing geometry may in particular be made from a material that is resistant to water vapor at temperatures of at least 120° C., particularly preferably of at least 150° C. In particular, it is conceivable for the sealing geometry to comprise a fluorine-containing rubber that is water-vapor-resistant and high-temperature-resistant in such a way.

(27) Alternatively or additionally, the maintenance costs can be reduced further and/or the service life of the sealing geometry can be increased when the sealing geometry, in particular the sealing

lip, has a coating. The coating may be a plasma coating. It may be glass-like or diamond-like. In general, it may have a Mohs hardness greater than the rest of the material of the sealing geometry.

(28) The sealing geometry may be in the form of a multistage seal. To this end, it may have at least two sealing lips.

(29) Wear, in particular on the sealing geometry, can also be reduced in that the sealing geometry is configured to apply a pressure force that is as low as possible. If the sealing geometry has, for example, a radial shaft shaft sealing ring, the radial shaft shaft sealing ring may be designed to generate only a relatively low radial pressure force. The radial pressure force generated can be quantified by a circumferential radial pressure force. Preferably, the circumferential radial pressure force may be at most 200 N/m, particularly preferably at most 160 N/m. As a result of such a low radial pressure force, a friction-related increase in temperature in the sealing geometry is reduced or avoided. In spite of the relatively high water content of the lubricant, a formation of water vapor in the region of the sealing geometry can be avoided or at least reduced as a result. The material of the sealing geometry can thus be exposed to relatively little stress. This advantage can be particularly meaningful when the sealing geometry comprises a fluorine-containing rubber, in particular an FKM.

(30) To reduce or avoid the ingress of contaminants into the drive unit, a contaminant labyrinth seal may be arranged on or in the drive unit. Preferably, the contaminant labyrinth seal may be arranged in the exterior and/or adjoin the exterior.

(31) The scope of the invention also includes a mobile power tool, for example a hand-held power tool, in particular for working on natural or artificial, in particular cementitious, rocks, wherein the mobile power tool has a drive unit and a sealing geometry, wherein the sealing geometry is designed to seal off an interior, containing a lubricant, of the drive unit from an exterior outside the drive unit, wherein the mobile power tool, in particular the drive unit, has a sealing geometry according to the invention.

(32) The mobile power tool in which the sealing geometry is used can be configured as an electric hand-held power tool. It can also be designed as a construction robot, in particular designed for working in civil engineering above and/or below ground.

(33) It can be configured and/or usable as a drill, chiseling machine, hammer drill, as a direct setting device or the like. The mobile power tool can be a high-performance device, in particular with a consumed power of at least 1 kW, particularly preferably of at least 2 kW. In particular, the mobile power tool can be intended to be operated at powers that lie in the region of a maximum power able to be drawn from a power source, in particular a power grid or a power storage unit.

(34) As a result of the reduction in friction that is achieved, a notable increase in the work output of the mobile power tool can thus be achieved, compared with previous mobile power tools of the same performance class, in particular in the region of the maximum power able to be drawn from the power source.

(35) The drive unit may have at least one transmission. The lubricant may be designed to lubricate at least a part of the drive unit, in particular the transmission or at least a part of the transmission.

(36) Particularly preferably, the mobile power tool can be cordless. In cordless mobile power tools, the total energy that is available is usually limited. The work outputs of cordless mobile power tools are also often reduced compared with corded tools. As a result of the reduction in friction achieved according to the invention, operating lives can be lengthened. It is also possible for the previously reduced work outputs to be increased and raised for example to the level of corded tools, wherein, at the same time, the flexible use of cordless mobile power tools can be maintained.

(37) To this end, the mobile power tool can have a rechargeable power storage unit. For example, it can have a battery, in particular a lithium-based battery. Alternatively or additionally, it is also conceivable for the mobile power tool to have a fuel cell.

(38) The mobile power tool can preferably be designed to drive a diamond-containing tool. The diamond-containing tool can be for example a diamond drill bit, a diamond saw blade or the like.

(39) The sealing geometry can be configured to seal off a boundary region between a housing of the mobile power tool and a shaft driven by the drive unit, for example a shaft of a tool fitting of the mobile power tool.

(40) Alternatively or additionally, the maintenance costs of the mobile power tool, in particular the sealing geometry thereof, can be reduced and/or its service life extended further when the mobile power tool is configured such that the temperature of the sealing geometry remains below 120° C., preferably below 100° C., during operation of the drive unit. To this end, the mobile power tool can have a cooling device. If the sealing geometry has or is configured as a dynamic seal, meaning that it is designed to provide sealing at a part that moves relative thereto, for example at the shaft, the sealing geometry can adjoin, in particular only, at least one part that moves relatively slowly relative thereto. A rotational speed of the at least one part that moves relative to the sealing geometry can be less than 10⁴ revolutions per minute, in particular less than 8×10³ revolutions per minute. The rotational speed can be at most two fifths, preferably at most one third, of the rotational speed of a previously conventional generic mobile power tool, for example of a mobile power tool having 22×10³ revolutions per minute. To this end, it is conceivable for example to enclose the drive unit in a housing. Only a shaft, for example for driving a tool fitting of the mobile power tool, can emerge from the housing. Such a shaft is usually located on an output side of the drive unit and therefore usually exhibits relatively low speeds.

(41) The lubricant can be an aqueous lubricant. It can have the features described above in connection with the sealing geometry.

(42) The scope of the invention also includes the use of a mobile power tool according to the invention for working on a natural or artificial rock, preferably for demolishing and/or erecting a building. In the case of such a use of the mobile power tool according to the invention, particularly high work outputs are frequently necessary, and so the value of the invention can be particularly pronounced in the case of this use.

(43) Further features and advantages of the invention emerge from the following detailed description of exemplary embodiments of the invention, with reference to the figures of the drawing, which shows details essential to the invention, and from the claims. The features shown there are not necessarily to be understood as true to scale and are shown in such a way that the special features according to the invention can be made clearly visible. The various features can be implemented individually in their own right or collectively in any combination in variants of the invention.

(44) In the schematic drawing, exemplary embodiments of the invention are shown and explained in more detail in the following description.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) In the figures:

(2) FIG. 1 shows a hand-held power tool;

(3) FIG. 2 shows a schematic illustration of a detail of a drive unit of the hand-held power tool having a sealing geometry and having lubricant.

DETAILED DESCRIPTION OF THE INVENTION

(4) In order to make it easier to understand the invention, the same reference signs are used in each case for identical or functionally corresponding elements in the following description of the figures.

(5) Although the invention generally encompasses mobile power tools and therefore for example construction robots or hand-held power tools, the invention is explained using the example of a hand-held power tool, only to make it easier to understand.

(6) FIG. 1 shows a hand-held power tool **10**. The hand-held power tool **10** is configured as a

hammer drill. It is cordless. To this end, it has a rechargeable battery **14** in the region of a housing **12**. The battery **14** exhibits lithium. The hand-held power tool **10** is configured as a portable device. It preferably has a weight of between 0.5 and 15 kg and generally of less than 25 kg.

(7) The hand-held power tool **10** also has a tool fitting **16**. A tool **18** is held in the tool fitting **16**. The tool **18** has a diamond drill bit.

(8) In a schematic illustration, a drive unit **20** of the hand-held power tool **10** is also discernible in FIG. **1**. The drive unit **20** is located inside the housing **12** and is illustrated in a manner superimposed on the housing **12** only for reasons of illustration.

(9) The drive unit **20** drives a shaft to which, in turn, the tool fitting **16** is coupled.

(10) The drive unit **20** has an electropneumatic impact mechanism and a rotary drive, which drive the shaft in a striking and rotating manner, respectively. The impact mechanism and the rotary drive are mechanically connected via a transmission of the drive unit **20** to an electric motor of the drive unit **20** and are able to be driven thereby.

(11) FIG. **2** shows a detail of the drive unit **20** in a schematic cross-sectional view.

(12) In particular, a sealing geometry **22** is apparent, which is designed to provide sealing between a shaft **24** driven by the drive unit **20** and the housing **12**, which is not illustrated in detail in FIG. **2**. In particular, the sealing geometry **22** seals off an interior IB, containing a lubricant **26**, of the drive unit **20** from an exterior AB. In FIG. **2**, the lubricant **26** is indicated schematically in the form of a plurality of wavy lines. In this case, the lubricant **26** can in principle reach the entire free and continuous space within the interior IB and spread out within this space.

(13) Ideally, the lubricant **26** cannot leave the interior IB.

(14) The lubricant **26** is an aqueous lubricant. It contains 33 percent by weight water. Thus, the hand-held power tool **10** (FIG. **1**) has the drive unit **20**, wherein the drive unit **20** has the aqueous lubricant **26**.

(15) As will be explained in more detail below, the sealing geometry **22** is designed to be used to seal off the aqueous lubricant **26** and as a result is particularly suitable therefor.

(16) The sealing geometry **22** is configured to radially encircle the shaft **24**. In particular, the sealing geometry **22** and the shaft **24** are configured symmetrically to the longitudinal axis of the shaft **24**.

(17) The sealing geometry **22** has a seal carrier **28**. The seal carrier **28** can be made from NBR. The seal carrier **28** is fixed to the shaft **24** by way of a press fit. Alternatively or additionally, the seal carrier **28** can be vulcanized onto and/or adhesively bonded to the shaft **24**.

(18) On its side facing the rest of the interior IB, the seal carrier **28** has at least one vortex structure **29** for creating at least one vortex flow within the lubricant **26**. In particular, the seal carrier **28** has two vortex structures **29**. The vortex structures **29** are configured in a channel-like manner. Their cross section can be configured in an elliptical, in particular circular, or at least substantially elliptical manner. The housing **12** can have a vortex structure configured so as to be complementary to at least one of the vortex structures **29**. By way of the vortex structures **29**, during operation of the drive unit **20**, one or more vortex flows can form in lubricant **26** flowing toward the sealing geometry **22**. In particular, vortex flows in opposite directions can form in the region of the vortex structures **29**, as are marked by circular-arc arrows in FIG. **2**. As a result of these vortex flows or as a result of the vortex structures **29**, the sealing action of the sealing geometry **22** can therefore be further improved.

(19) Fitted in an indentation **30** in the seal carrier **28** is a sealing lip **32**. The sealing lip **32** has a V-shaped cross section. Its sealing lip wing **34** projects from the seal carrier **28** obliquely, in particular laterally, i.e. in a direction parallel to the longitudinal axis of the shaft **24**, and in a radial direction relative to the shaft **24**.

(20) The sealing lip **32** is made from an elastomer. In one exemplary embodiment of the invention, to this end, the sealing lip **32** is made from NBR. In an alternative exemplary embodiment, the sealing lip **32** is made from a fluorine-containing rubber, in particular from an FKM. The sealing

lip **32**, in particular the sealing lip wing **34**, can have a plasma coating, in particular for reducing the friction of the sealing lip wing **34** against the housing **12**.

(21) When the drive unit **20** is at a standstill, the free end of the sealing lip wing **34** and thus the sealing lip **32** bears laterally against the housing **12**. In this respect, the housing **12** forms a mating surface for the sealing lip **32**. In particular, the sealing lip wing **34** presses against the housing **12** with a, preferably low, pressure force. Therefore, when the drive unit **20** is at a standstill, the sealing lip wing **34** seals off the interior IB, which is delimited from the exterior AB by the housing **12** and the seal carrier **28**, inter alia, from the exterior AB. The lubricant **26** located in the interior IB cannot escape into the exterior AB.

(22) During operation of the drive unit **20** and therefore during rotation of the shaft **24**, the sealing lip wing **34** is deformed in a direction radially away from the shaft **24** on account of centrifugal force. As a result, the pressure force of the sealing lip wing **34**, with which the latter is pressed against the housing **12**, is reduced. If the speed of the shaft **24** exceeds a minimum rotational speed, the sealing lip wing **34** is lifted at least slightly off the housing **12**.

(23) The friction between the sealing lip wing **34** and the housing **12** and therefore the otherwise locally arising relative temperature increase are, as a result, diminished or avoided. A local formation of water vapor in the region of the sealing lip wing **34** can therefore likewise be reduced in scope or even avoided entirely.

(24) The sealing action of the sealing geometry **22** is maintained even in the event of the sealing lip wing **34** lifting off on account of the resultant centrifugal forces, as a result of which lubricant **26** flowing up is forced out of the boundary region between the seal carrier **28** and the sealing lip **32** and the housing **12**.

(25) In order to additionally support this dynamic sealing effect, a lubricant labyrinth seal **36** having a constriction **38** in the form of a narrow channel is formed between the housing **12** and the seal carrier **28**.

(26) At least one thread can be formed in the region of the lubricant labyrinth seal **36**, in particular in the region of the constriction **38**. By way of the thread, lubricant **26** can additionally be conveyed away from the sealing lip **32** during operation of the drive unit **20**. To this end, the thread can preferably be formed so as to extend in a radial direction or at least so as not to extend axially with respect to the shaft **24**.

Claims

1. A sealing geometry for a mobile power tool wherein the mobile power tool has a housing forming a mating surface, and a drive unit, wherein the sealing geometry is designed to seal off an interior (IB), containing a lubricant, of the drive unit from an exterior (AB) outside the drive unit, the sealing geometry having a seal that is contactless at least during operation of the drive unit, wherein the sealing geometry has a sealing lip sealing against the mating surface arranged at the housing of the mobile power tool.
2. The sealing geometry as claimed in claim 1, wherein the sealing lip bears against the mating surface of the mobile power tool when the drive unit is at a standstill, in order to seal off the interior (IB) from the exterior (AB).
3. The sealing geometry as claimed in claim 2, wherein the sealing geometry has a seal carrier on and/or in which the sealing lip is arranged.
4. The sealing geometry of claim 3, wherein the sealing lip fits in an indentation in the seal carrier.
5. The sealing geometry as claimed in claim 2, wherein the sealing lip has a V-shaped or an at least substantially V-shaped cross section.
6. The sealing geometry as claimed in claim 1, wherein the sealing lip is configured such that, during operation of the drive unit, the sealing lip comes away from the mating surface of the mobile power tool or that, during operation of the drive unit, a pressure force of the sealing lip on

the mating surface is reduced at least compared with when the drive unit is at a standstill.

7. The sealing geometry as claimed in claim 1, wherein the sealing geometry is designed to create at least one vortex flow in the lubricant) during operation of the drive unit, said vortex flow improving the sealing action of the sealing geometry.

8. The sealing geometry as claimed in claim 1, wherein the sealing geometry has a seal carrier having at least one vortex structure for creating the at least one vortex flow.

9. The sealing geometry as claimed in claim 1, wherein the sealing geometry is designed to have and/or to create a sealing fluid at least during operation of the drive unit.

10. The sealing geometry of claim 9, wherein the sealing geometry is designed to have and/or create a sealing air at least during operation of the drive unit.

11. The sealing geometry as claimed in claim 1, wherein the sealing geometry has a lubricant labyrinth seal.

12. The sealing geometry as claimed in claim 1, wherein the sealing geometry comprises an acrylonitrile butadiene.

13. The sealing geometry as claimed in claim 1, wherein the sealing geometry comprises a fluorine-containing material.

14. The sealing geometry as claimed in claim 1, wherein the sealing geometry comprises a water-vapor-resistant material.

15. The sealing geometry as claimed in claim 1, wherein the sealing geometry comprises a high-temperature-resistant material.

16. The sealing geometry as claimed in claim 1, wherein the sealing geometry, has a coating.

17. A mobile power tool wherein the mobile power tool has a housing and a drive unit and a sealing geometry, wherein the sealing geometry is designed to seal off an interior (IB), containing a lubricant, of the drive unit from an exterior (AB) outside the drive unit, wherein the mobile power tool has the sealing geometry according to claim 1.

18. The mobile power tool as claimed in claim 17, wherein the lubricant is aqueous.

19. The mobile power tool as claimed in claim 18, wherein the lubricant comprises, in addition to water, at least one additive.

20. The mobile power tool as claimed in claim 17, wherein the lubricant comprises at least 5 percent water.
